

Perceptions of Gender, Mobility, and Personal Safety: South Africa Moving Forward

Transportation Research Record
2019, Vol. 2673(11) 616–627
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DOI: 10.1177/0361198119854087
journals.sagepub.com/home/trr


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Abstract

Transport users make mode choices based on a variety of factors. These factors are economic or service driven, based on individual roles, habits, and interests, as well as age, life cycle stage, and gender. Analysis reflects different mobility patterns for males and females relating to care activities. Literature suggests that experiences of harassment have a significant effect on user choices. This study examines how South African data compares with international studies. Mode use and trip purposes, distances, and times differ depending on gender and are affected by the experience of harassment, which affects females more than males. Analyzing trip making in South Africa revealed that travel modes, distances, and times are not significantly different across gender. South African females make fewer trips than males, but significantly more care trips. Different modes of public transport score differently regarding potential experience of harassment, with trains performing the worst. The reason mode choice is not significantly different between females and males is assumed to be because of financial reasons. Investigating harassment perception in Cape Town reveals that females experience harassment more often and this influences their choices regarding care trips. These findings have significant implications for transport policy in South Africa and suggest that more nuanced policies are required.

The South African Constitution (1) states that “everyone has the right of freedom of movement” and that “every citizen has the right to enter, to remain in, and to reside anywhere in the Republic.” It also prescribes that the state may not unfairly discriminate, directly or indirectly, against anyone on one or more grounds, including gender.

Civil rights of citizens within urban areas are dynamic, responding to that environment: “The right to the city is far more than the individual liberty to access urban resources: it is a right to change ourselves by changing the city” (2). This implies that the right to the city offers hope for change, not only for future cities and their governance, but also for our “socially constructed” selves, so that women and men, girls and boys can live everyday lives in both recognition and equality (3).

Transport as a sector is assumed to be gender neutral. Scrutiny of the access of women and men to public and private transport internationally, however, reveals that mobility patterns of men and women are different. Research into the field of transport planning and practice has consistently failed to apply a social science perspective in determining patterns of transport and travel,

including the demonstration of a systematic lack of understanding of the gender-based differences in system requirements resulting in mobility deprivation (4, 5). Mainstream transport planning still remains largely untouched by debates on diversity and gender differences (3). In the context of the global South, research on gender and transport issues is still relatively scarce (6).

The South African National Development Plan (7) sets a positive framework in which to position specific policy initiatives. The vision is that, by 2030, investment in the transport sector will enable:

- Improved access to economic opportunities, social spaces, and services by bridging geographic distance affordably, reliably, and safely.

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- Economic development, by supporting the movement of goods from points of production to where they are consumed, facilitating regional and international trade.
- Greater mobility of people and goods through transport alternatives which support minimized environmental harm.

Here the state is positioned as the navigator within a framework of strong and effective regulation of public and private transport. The National Development Plan (7) recommends focusing on safe, affordable, and effective transport options to enable equitable access to opportunities for all and to reduce poverty.

Transport policy in South Africa is developed according to the White Paper on National Transport Policy 1996 (8), which sets out to provide a basis for transport to play a more strategic role in social development and economic growth to “promote a strong, diverse, efficient and competitive industry” (8) for the “development of a stable institutional capacity to manage the implementation” (8).

The objectives set out in the 1996 White Paper include:

- To support the goals of the prevailing, overarching plan for national development to meet the basic accessibility needs of the residents of South Africa, grow the economy, develop and protect human resources, and involve stakeholders in key transport-related decision making.
- To enable customers requiring transport for people or goods to access the transport system in ways that best satisfy their chosen criteria.
- To improve the safety, security, reliability, quality, and speed of transporting goods and people.
- To improve South Africa’s competitiveness and that of its transport, infrastructure, and operations through greater effectiveness and efficiency to meet the needs of different customer groups better, both locally and globally.
- To invest in infrastructure or transport systems in ways that satisfy social, economic, or strategic investment criteria.
- To achieve the above objectives in a manner that is economically and environmentally sustainable and minimizes negative side effects (8).

These goals form part of the structure of the revised White Paper published in 2017. It is out of these objectives that the focus on care mobility must be approached—specifically looking at harassment as an issue affecting the supply of safe, secure, and reliable transport within South Africa. To make sensible changes

to South Africa’s policies and practices, a good understanding of mobility differences between females and males is required. Do females make more (care) trips? Which modes of transport are used and what considerations influence these mode choices? This paper will unpack what these differences mean for care mobility in South Africa.

This paper comprises of two parts. In the first part, international trends in travel-pattern differences between females and males are introduced as the context for the research by referring to the literature. Secondly, the research examines available data on mobility, based on the South African National Household Travel Survey 2013 (SANHTS) (9), and different trends in mobility between females and males in South Africa. In particular, it looks for trends regarding trip frequency, duration, distance, and mode choice. Globally, it has been established that mode choice by females is influenced by their experiences of harassment and violence. The second part of this paper, therefore, unpacks harassment experiences in the South African context in an exploratory manner, which is based on primary data collected in Cape Town.

Methodologies

Data from the SANHTS (9) is examined to see how it engages with issues of harassment. It is evident that the SANHTS (9) only collects and provides information regarding harassment provided by household heads within the home, either female or male, rather than household members’ experience of harassment in their utilization of transport options.

The secondary data, provided by Statistics South Africa, was complemented by primary data acquired by fieldwork with four focus groups from particular constituencies and interviews with individual rail commuters.

For the focus groups, four group discussions were conducted in 2015 and 2016, with between seven and ten participants each. The groups were selected within particular urban areas representative of typical communities making choices regarding transport on a regular basis.

The four groups were:

- A group of unemployed females from Langa, an established township within the greater Cape Town municipality.
- A group of general female bus users.
- A female group from Atlantis (a satellite town).
- A male group (viewed as the control) from Atlantis.

Notes were taken during the focus groups relating to reasons for transport mode choice as well as experiences relating to harassment. This involved both anecdotes

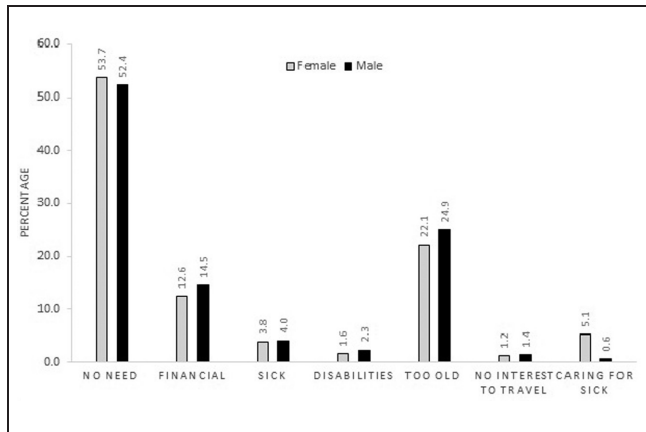


Figure 1. Reasons for not traveling in South Africa (past seven days) (9).

with regard to personal experiences and experiences relating to persons known by the interviewee (secondhand). These findings were supplemented by interviews with 285 rail passengers. Rail passengers were interviewed on the station platform regarding their experiences of harassment and their answers were recorded on paper.

Mobility Patterns of Women and Men

The International Context

With limited access to individual means of transport, the vast majority of female residents in developing countries are dependent on either walking or on public means of transport, which can be both motorized (e.g., buses, subways) or non-motorized, such as rickshaw taxis in Asia, or boda-boda and bicycles in Uganda (10). Women tend to use different and cheaper modes of transport than men, particularly women with low incomes (3).

As reported in literature from different parts of the world, adult women make more person trips than men of the same age group: see, for example, McGuckin and Murakami (11). In the U.S.A. over 60% of married women are part of the paid labor force, while still retaining substantial child care and domestic obligations (11). Women typically have to make multiple stops, pay multiple fares, and travel during off-peak hours when public transport services are less reliable and waiting areas are less safe (10). Vance and Iovanna (12), also in the U.S.A., established that gender differences are more distinct among younger individuals.

Gender-blind provision of urban services may not meet the needs of women if their priorities are not taken into consideration (13). Poor women and men tend to walk more than those who are better off, with low-income women tending to walk the most (14, 15).

In examining data for a different context, referencing Spain, Negrón-Blanco et al. (16) established the housebound status (people who do not travel) as 9.8% for the Spanish population. Housebound individuals tended to be older, female, and having a lower educational level (16).

Internationally, travel time expenditure is strongly related to individual and household characteristics, including gender (17). The literature on gender and travel time shows varied results. A number of researchers have found that men spend more time traveling than women (18–23). Lu and Pas (24) found that women spent more time traveling than men; however, they acknowledged under-reporting of short trips, which might be more commonly made by women, affecting the validity of their conclusion. An insignificant relationship between gender and travel time expenditure was found by Zahavi and Talvitie (25). Around the world, men predominantly use private vehicles and public transport to commute to work. Women, in contrast, make more frequent and shorter trips and stop more often. They are more likely to “trip chain” and to combine the daily commute with shopping, picking up or dropping off children, visiting relatives, or running small errands (26). Overall, it can be concluded that there are significant gender inequities globally.

The South African Context

This section elaborates on travel patterns in South Africa and, particularly, gender differences. The analysis is based on the SANHTS (9). The data collection included a total of 51,341 households (dwelling units), representing the South African total of 14.55 million dwelling units. The SANHTS provides weightings so that findings can be converted to the total South African population. At the time, South Africa had a population of 52.72 million people, of whom 49.9% were female. Of the 42.4 million people who took one or more trips during the seven days prior to the survey day, approximately 85% consisted of individuals in urban and metropolitan areas, while 75.4% were individuals living in rural areas.

Overall, 5.57 million women and 4.12 million men did not undertake any trips during the surveyed seven-day period. Figure 1 shows the reasons for not traveling. The majority (over 50%) of people who did not travel in the seven-day survey period indicated that they did not have a need/reason to travel (see Figure 1). The second most important reason for not traveling during the survey period is age related. Over 22% of women and almost 25% of men indicated that they were too old to travel. A total of 12.6% of females and 14.5% of men indicated that they did not travel because of financial reasons. Although care for children or the sick is not among the most important reasons for not traveling, the percentage of females who

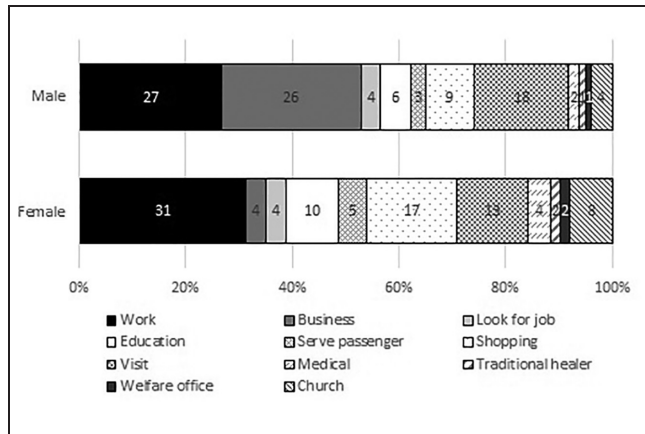


Figure 2. Trip purposes per gender in South Africa (%) (9).

stayed at home for this reason was significantly higher than their male counterparts—5.1% vs. 0.6%.

Based on the SANHTS (9), excluding the part of the population that did not make any trips during the survey period, women appear to make 2.3 trips on average per day, while their male counterparts make 2.9 trips. This is in contrast to international trends which show that women make more trips than men.

On a daily basis women need to balance competing demands which influence their daily travel. In South Africa (see Figure 2), a higher percentage of women than men (31% vs. 27%) travel to work, while males make more business trips (25% vs. 4%). Around 4% of trips made by both men and women are to look for a job. More women than men (10% vs. 6%) make educational trips (excluding children aged under 18 years). In line with international literature, South African women make more care trips. In relation to Figure 2, the categories “serve passenger” (taking others to school, clinic, etc.) and “shopping” represent care trips. Women make 2% more “serve passenger” trips and 8% more shopping trips than men, which indicates that South African women, like their international counterparts, make more care trips than men.

Internationally, travel experiences between men and women vary (18–23, 26). Figure 3 shows the modal split according to gender in South Africa.

The most prevalent mode in South Africa is walking: 48.6% of females and 47.6% of males walk for all trip purposes. More males (16.9%) than females (11.8%) are car drivers, while more females (13.2%) than males (10.9%) are car passengers. Slightly more females (26.5%) than males (23.5%) use public transport. Females favor bus and minibux taxi (MBT) over train travel.

Gender disparities in relation to transport need to be investigated in the light of earning power, because men

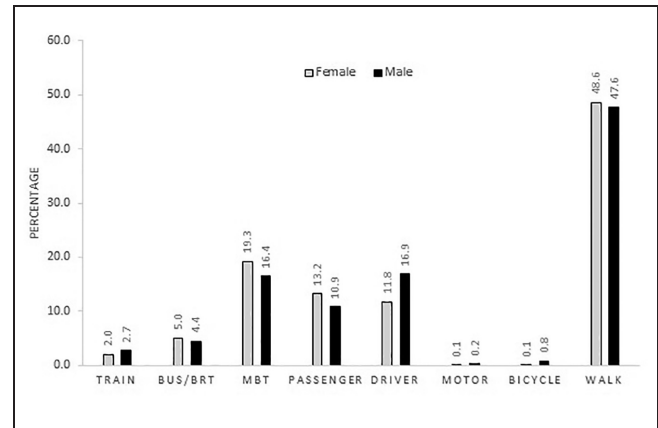


Figure 3. Modal split per gender in South Africa (%) (9).

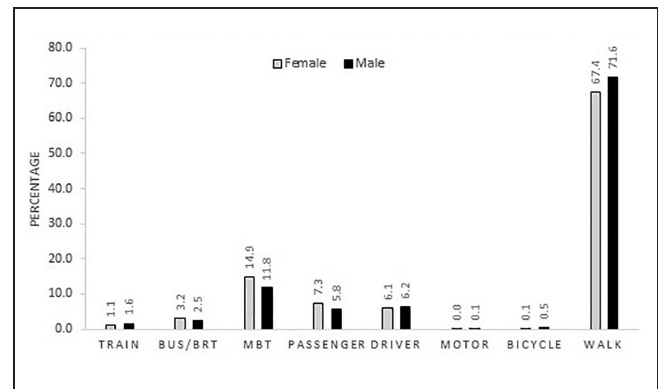


Figure 4. Modal split per gender for the poorest quintile in South Africa (%) (9).

and women experience and respond to financial strains differently. This affects travel volumes and mode choices. Access to income and assets, housing, transport, and other basic services is influenced by gender-based constraints and opportunities. As mentioned, low-income women and men tend to walk more than those who are better off, with low-income women tending to walk the most (14, 15).

Overall, 60% of trips made in South Africa by persons from the poorest income quintile are made by men, based on the SANHTS (9). The modal share between females and males does not differ substantially (see Figure 4). Less than 1% of females and males use bicycles and motorcycles. Train use is also very limited, however; slightly more males use this mode. Females are more inclined to use bus/bus rapid transit (BRT) or MBT, or travel as a passenger in private transport. There is no significant difference between the percentage of female and male drivers, while men are more likely to walk.

Compared with the overall population, those in the poorest quintile walk significantly more. In the case of

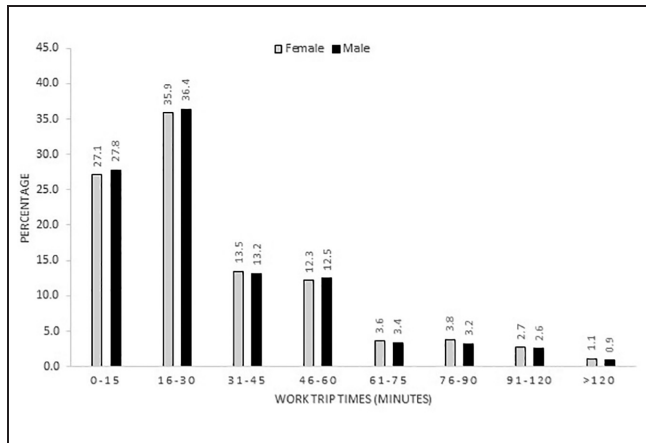


Figure 5. Travel time for work trips per gender in South Africa (%) (9).

females, walking increases by 38.6%, while male walking trips increase by 50.7%. Correspondingly, use of all other modes decreases among the poorest quintile, with car driver and passenger showing the most significant decline.

The literature indicates that there may be a relationship between gender, employment status and travel time expenditure. Prendergast and Williams (18) found that, internationally, a combination of gender and employment increased the differences between the maximum and minimum travel time significantly. The maximum average travel time expenditure by full-time employed males was about three times the minimum average travel time recorded for retired women.

SANHTS (9) only provides overall travel times for work trips and trips to educational institutions. Females travel for 35.5 min, on average, to an educational institution, while males travel for 34.6 min, on average. For the work trip, the averages are 46.4 min and 48.3 min, for females and males, respectively. The difference in overall travel time categories between females and males going to work is insignificant (see Figure 5); the maximum percentage difference per travel time category is 0.8%. Analysis of educational trips show similar results. Moreover, analysis of educational trips for adults (age 18 years and above) again shows insignificant differences. This contradicts international findings.

Personal Safety Experiences of Women and Men

The International Context

International literature identifies harassment as the indicator for personal safety in relation to public transport. All forms of harassment affect women more than men.

Women's access to opportunities and quality of life is particularly affected. Harassment can take various forms, such as verbal harassment (cat calling or unwanted teasing); visual harassment (such as leering or staring); and physical forms (for example, men exposing themselves, groping, or other forms of touching, etc.). Harassment often takes place in public places, as women travel to and from educational institutions or the work place. Harassment seems to occur especially in relation to public transport. This may be in, or around, bus and train stations, or other public transport hubs and stops, and on the vehicles themselves, especially if they are crowded (26).

Harassment is a problem for women on all transport modes and routes. For example, a study in Delhi, India, found that: "Women are the targets of sexual harassment while travelling to work and practically every woman interviewed had anecdotal evidence of suffering from the same. Harassment while walking down the street or travelling on a bus is a common occurrence for working women and is exacerbated by the absence of adequate lighting on streets and subways and by the small, lonely paths connecting the slum with the bus stops." (14).

Fear of crime is widely recognized as a barrier to public transport (27, 28). Research in the U.K. has identified that 10.5% more rail trips would be generated if people felt more secure when traveling and waiting at stations (27). A majority of car drivers in inner Los Angeles claimed they would use transit if public buses were perceived as safe and clean (29). The most recent survey by Eurobarometer of city dwellers in all European cities makes a correlation between life satisfaction and a "feeling of safety" (<http://ec.europa.eu/commfrontoffice/publicopinion/index.cfm>). Feeling safe while traveling around a city has effects on the confidence, productivity, and quality of life of everyone (26), but more particularly women.

Allen and Vanderschuren (26) established that women face harassment while using public transport on a daily basis all over the world. It is reported as an issue of concern from Baku, Azerbaijan to Bogotá, Colombia—and everywhere in between. It is also not confined in any way to the developing world, and appears to be just as prevalent in rich, highly industrialized cities, such as London, New York, or Paris.

Acts of harassment happen frequently on public transport. Women use public transport that is overcrowded and travel for long distances. They are most likely to experience harassment, be it verbal or physical, from the men using the same mode of transport. Overcrowding is a high security issue for women, as it is the breeding ground for physical harassment and inappropriate behavior (26). Women are also concerned about their security on the journey to access public transport, as they may

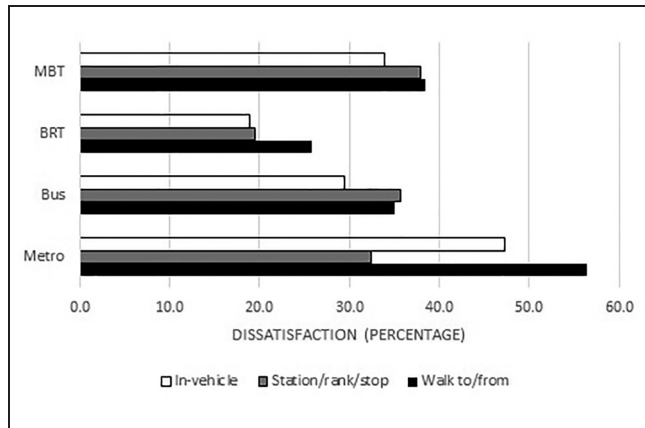


Figure 6. Security concerns per mode according to household heads in South Africa in 2013 (%) (9).

have to walk along the road if there is no safe sidewalk, or in the dark if there is no street lighting. They may also be fearful of unwelcome advances from others, as well as the journey in the public transport vehicle and, of course, have the same concerns for their trip to their final destination.

The South African Context

Between the 2003 (30) and 2013 SANHTS (9), access to private motorized transport (i.e., car ownership per household) in South Africa increased from 25% to 35%. This means that the majority of people still have no access to private vehicles and, therefore, have to use (formal or informal) public transport or non-motorized transport. Household heads questioned for the SANHTS in 2013 reported that personal safety is a major concern. Figure 6 provides an overview of the reported harassment perception in South Africa for various public transport modes.

An alarming percentage of household heads expressed concerns related to personal safety related to Metrorail train services (see Figure 6). Metrorail, operated by the Passenger Rail Agency of South Africa (PRASA), is a commuter rail service in the major urban areas of South Africa. Metrorail is responsible for 2.2 million trips daily (38% at the Witwatersrand/Gauteng, 30% in the Western Cape, 15% in Tshwane/Pretoria, and 13% in Durban). Household heads were particularly concerned regarding the walking trips to and from the station (56.5%). Many also expressed concern regarding the part of the journey spent on the train (47.3%), and 32.4% indicated concerns regarding the Metrorail stations.

For all parts of the trip with MBTs and buses, around 30% of household heads indicated concerns with personal safety. Despite BRT being perceived as the most secure mode of public transport, around 20% of

household heads stated security concerns regarding this mode. Given the fact that various South African metropolitan areas are investing in BRTs, this statistic is still considered high. Household heads reported that they were (very) unsatisfied with the walk to/from public transport (35.6%), waiting for the public transport vehicle (27.7%), and while inside the public transport vehicle (27.8%).

Focus Group Discussions in Cape Town. The results of four focus group discussions are provided in Table 1. All the respondents were of working age (between 18 and 61 years).

The mobility of the unemployed females focus group was relatively low (they traveled out of the township less than one day per week, and often only once a month), compared with their working counterparts, who on average traveled from five to seven days a week. The group of unemployed females traveled to collect their welfare grant once a month and did their grocery (and other) shopping at the same time. Their working counterparts traveled mostly for work and leisure. The female group from Atlantis also traveled for medical and grant collection purposes.

When respondents travel alone, especially female passengers, they often meet up with the same people while traveling. Over time, people will start keeping an eye out for each other. If they are planning not to travel the next day, they will let other passengers know.

The unemployed females, on average, used the train because it is the cheapest option. Their employed counterparts used the bus, which is deemed better and relatively safe. The BRT system is safer, but expensive (and overcrowded in some cases).

There is a clear difference regarding the perception of risk between males and females. Male bus users in Atlantis did not perceive a security risk during their trip at any time of the day, while females in Atlantis did not like to travel in the evening. Although they did not use the train, they perceived the off-peak hours, when it is quiet, as dangerous. The other female focus groups had a higher risk perception. They thought that public transport modes are dangerous at any time of the day. While males found the trip to the station/stop/rank the most dangerous, females were also intimidated and felt insecure waiting at the platform/stop and in the train.

Remedial action varied widely between the focus groups. The unemployed females indicate that they did not have solutions (although the presence of neighborhood stewards did help at certain times of the day). This feeling of heightened insecurity may also be associated with the fact that they made few trips outside of their habitual areas and, therefore, were less comfortable in the public transport environment. The working bus users

Table 1. Cape Town Focus Groups Findings

Criteria	Langa unemployed females	Cape Town bus females	Atlantis bus females	Atlantis bus males
Sample size	10	8	10	7
Age range	18–60	37–54	43–61	23–60
Employment (%)	0%	100%	50%	57%
Number of days travel/week	<1	5	7 (1)	7 (7)
Reason for travel	Collect social welfare grants, shop at same time, hospital or clinic	Travel to work, other services, shopping and leisure	Travel to work, hospital/clinic, collect social welfare grant, shopping and leisure	Travel to work, shopping and leisure
Travel companions	Alone or with friend/s	Alone, with same passengers daily	Friend/s, spouse	Wife
Mode choice	Paratransit Bus BRT Train	Not preferred Better option Expensive Not preferred	Unavailable Better option Expensive, overcrowded Station too far	Expensive Only option Expensive Station too far
Security questions Feeling toward public transport	Paratransit	Violence between operators	Not safe	Relatively safe
Dangerous times to use public transport	Bus BRT Train Paratransit	Relatively safe Safer than other modes Least safe Any time, sexual harassment happens more during peak	Relatively safe Safer than other modes Least safe After work	Relatively safe Safer than other modes Least safe None
Where does harassment take place?	Bus BRT Train	On route to bus stop, on platform, and on the train	In the evening None Between 09:00 and 16:00 hours	On route to bus stop
Reason for harassment Remedial action for violence	Unknown Nothing seems to help; local stewards to accompany students, school children, and women to the stations and home have helped	No respect Leave valuables at home; tell bus company	No respect Tell conductor and other passengers	Women seen as weak Raise awareness about violence

Note: BRT = Bus Rapid Transit; Paratransit = informal transport which operates between public and individual private spheres.

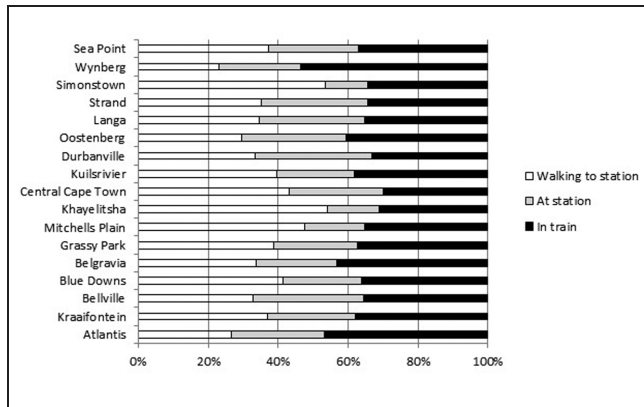


Figure 7. Risk perception related to train travel in Cape Town (%) (9).

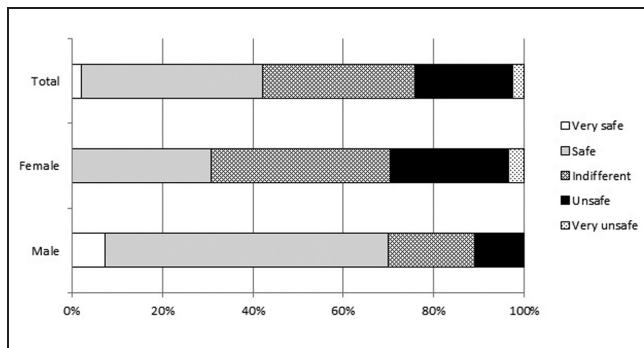


Figure 8. Personal safety according to interviewed train passengers ($n = 285$).

would leave their valuables at home, minimizing the risk of theft. In Atlantis, females shared their experiences of harassment with the conductor and other passengers, while males would raise awareness about violent behavior among bus passengers and within the community. In the focus groups, almost all women shared different strategies regarding how they coped with the personal security threat and harassment. This included not traveling at night or in the dark, only taking a small amount of money, and using the phone (or pretending to speak to someone).

Train Passenger Interviews in Cape Town. The general perception of the focus groups was that the train poses the highest security risk but, when the security risk of train travel is disaggregated, the type of perceived security risk differs. More than 10% of respondents had personal fears about being at the station, while the station is perceived to be less dangerous than walking to/from the station or being on the train. Using Cape Town traffic analysis zones (TAZs), for example, in areas such as Simonstown,

Central Cape Town, Khayelitsha, Mitchells Plain, and Blue Downs, over 40% of respondents feared walking to the station most. In Wynberg, Oostenberg, Belgravia, and Atlantis, more than 40% of respondents perceived the train ride itself as the most dangerous part (see Figure 7). Other public transport modes show similar variations.

As indicated above, Simonstown is one of the areas identified as having train-related security issues. In addition, many train users in this area are choice users (in that they have access to cars as an alternative mode of transport). The study team, therefore, decided to interview train users in Simonstown, focusing mainly on female travelers, in line with the research topic. However, for comparison reasons (control group), some male travelers were also questioned. In total 202 females (70.9%) and 83 males (29.1%) were interviewed. Of these travelers, 80.3% used the train as their main mode of transport throughout the year. Some 27.7% of travelers traveled Monday to Friday, while another 10.9% traveled Monday to Saturday, and 7.4% traveled every day of the week. The remainder of travelers traveled less than five days: 9.9% traveled four days a week and 23.5% three days a week.

Based on the SANHTS (9), when using public transport, a total of 91.6% of users in Cape Town indicated that they favor train travel, mainly because it is cheaper than other public transport options (75.1%). Almost 65% of travelers are alone when they travel. For the people who travel with a companion, most travel with their spouse, children, other family (e.g., brother/sister), or friends.

Train users in Simonstown were asked how secure they feel when using the trains (see Figure 8). The vast majority of the males interviewed (89.2%) felt that train travel does not pose a security threat to them, or they were indifferent about their personal security in train travel. A majority (70.3%) of female interviewees also felt secure, or indifferent. However, there was a large difference between the numbers of females (30%) and males (70%) who felt safe or very safe—no females felt train travel was very safe. It needs to be pointed out here that the interviews for this study were conducted on the southern suburban line—a line that is considered more secure than other lines in the Cape region.

Train users were also asked how secure they feel during the ingress and egress part of the train journey. It was found that 87.5% of males and 84.4% of females felt less secure when traveling to/from the station. Nevertheless, this generally does not keep them from using the train: 84.1% of males and over 90% of females indicated that they would use the train regardless of the security risk.

When asked if the train users themselves, or any of their family members, have ever been harassed using the

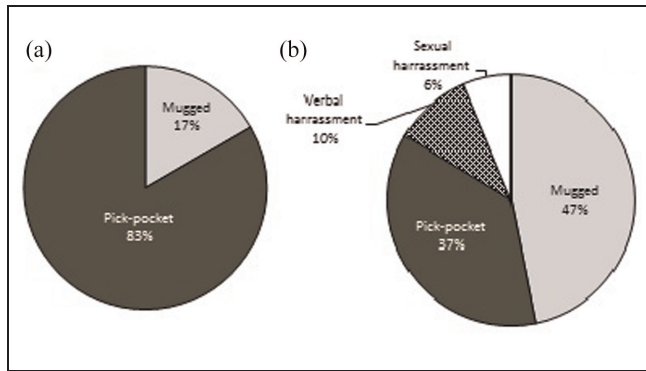


Figure 9. Types of harassment for (a) males ($n = 24$) and (b) females ($n = 79$).

train, 28.9% of males and 37.6% of females indicated that they had been harassed or have a family member who had been harassed.

There is a clear difference between the harassment that men and women experience, according to the train users interviewed in Simonstown (see Figure 9a and b). Both males and females were worried about petty crime, such as mugging (males, 17%) or pick-pockets (males, 83%) on (the way to/from) the train. Females experienced more than double the rate of muggings (47%), but less pick-pocketing (37%). They also encountered verbal harassment (10%) and sexual harassment (6%). The majority of males (88.8%) and females (81.6%) indicated that the harassment happens mostly on the train. Neither males nor females indicated that robbery or harassment is acceptable.

When the train users were asked what measures should be put in place to combat security issues, the overwhelming comment was the need for more security personnel. A total of 82% of males and 87% of females indicated that more security personnel are required. Males did not indicate any other potential measures, while females suggested that people can ask for help (3%) and that workshops with criminals could be held (2%). The workshops should be focused on the youth, as they are the main cause of the harassment on the trains, according to these respondents.

Train users do not think that harassment is encouraged by travellers: 95.8% of males and 93.2% of females indicated that this is not the case. When asked if the train users ask for help if they are harassed, some 58.8% of males indicated that they do and 88.8% of them would ask other train users for help. Only 25% of males indicated that they do report crimes, however, of the males that do report crimes, 95.5% indicated they had confidence in the reporting channels.

Of the female train users, 81.3% indicated that they would ask for assistance, of which 75% would ask other

train users, while 18.8% would contact the transport operator (Metrorail) and 6.3% would contact the police. A total of 57.9% of female train users indicated that they actually report the crimes. In contrast to the male respondents who are very confident about the reporting channels, only 27.4% of females indicated confidence in the reporting channels.

The Cape Town train users on the southern suburban line were aware that crime is also an issue in other cities. All females indicated that this is the case, while 95.5% of males indicated the same.

Conclusions, Recommendations, and Policy Implications

South African mobility patterns show both similarities and significant differences to international trends. The phenomenon of gender-related travel differences is the topic of numerous studies (10–12, 16), but the findings in the literature are not always significant and can contradict each other. This research project sets out to supplement the quantitative data obtained in large samples with qualitative data of smaller samples, obtained through more in-depth focus group interviews.

Internationally, one gender pattern that has been identified relates to the tendency to stay home and avoid travel being more prevalent among females (16). This correlates to the findings in this study, although the percentage of people who are housebound in South Africa is larger, around 20% (9), compared with around 10% internationally (16). Although care for children or the sick is not among the most important reasons cited for not traveling (9), the percentage of females who stay at home for this reason is significantly higher than their male counterparts, 5.1% vs. 0.6%. Internationally, females make more trips than their male counterparts, mainly because of the additional care trips that females make. In South Africa trip making for males is higher, although females do make more care trips (2% more “serve passenger” trips and 8% more shopping trips), which is in line with international literature.

Interestingly, South African females and males do not have significant differences in travel distances, travel times or (to a lesser extent) mode choice. This contradicts international studies where differences are often found. The lack of significant gender-related differences between mode choices is remarkable, as the perception of harassment varies significantly between females and males. Overall, between 20% and 56.5% of household heads consider (parts of) the public transport journey risky. Trains are considered the least safe and are mainly used for affordability reasons. Most dangerous is the walk to/from the station, followed by the time spent on the train. The station itself is considered relatively safe.

The focus group discussions revealed that users of MBT and buses, especially female travelers, combat harassment risks by traveling in pairs or with the same passengers every day. The time of day does not appear to influence the risk of harassment significantly in the South African context, as revealed during the focus group discussions. This contradicts global findings, where evening travel was considered more dangerous (26).

The contribution by transport users of concrete and positive suggestions to improve safety in the transport sector through enhancing the visibility and deployment of security personnel demonstrates engaged, active citizens with a vision for improved service delivery. This augurs well for moving transport delivery in South Africa into a new and safer era.

Based on primary and secondary data, it can be concluded that there is gender inequity in the South African transport system, especially in the public transport services. The inequity is in sharp contrast with the country's progressive Constitution. Furthermore, treating communities as homogenous is a major flaw in current policies and practices. These policies fail to recognize that women and men have varying mobility and accessibility needs.

Currently, there is little guidance for transport planners and city authorities to plan better and more gender sensitive networks and services. There is significant need to develop more responsive and nuanced measures within transport modes, accommodating women's personal safety requirements, as gender-blind practices have been failing to address their needs.

The South African government has policies and strategies to invest in public transport modes as safe, affordable, and reliable systems, accessible to all. Recent investments in public transport have not resulted in the anticipated personal safety improvements. Consequently, the policy recommendations derived from this paper are:

- Improvements to the South African transport system require public transport integration providing improved personal safety. This will encourage women within South African cities to utilize a safer public transport network available to them, accommodating the care trips that women make in the service of others.
- Public transport and non-motorized transport masterplans should feature women's need for personal safety as a unifying element for investment. Key linkages are needed between communities and public transport stations/precincts and the required infrastructure. The infrastructure would include: proper lighting, secure paved sidewalks, and designs that encourage safety and security—especially around rail stations, as train travel is the transport mode where women feel most unsafe.
- Civil society should be actively involved in assisting government with awareness campaigns, with a focus on the personal safety of women in public transport and the options available within the law to address effectively the security concerns raised.
- The MBT industry remains unregulated within South Africa. With a focus on integrated public transport networks, the MBT is an integral part of the South African public transport system and has the opportunity to position itself as a safe and secure service. Government should partner with MBT organizations to advocate for better security at taxi ranks. Pilot projects are underway.
- Visible policing and security, at night and during the day, should be the norm for public transport precincts. Currently, this is common practice for BRT stations within South Africa and the perceptions is that BRT is the safest mode of public transport.
- The research paper has shown that men and women perceive safety differently in South Africa. The SANHTS (9) rationale of only addressing questions to the heads of households might be limiting. Thus, it is recommended that the SANHTS uses women within the household as the benchmark for security concerns.
- The country has the necessary legislative frameworks in place to ensure the safety of women in public transport. However, there is a lack of adequate implementation (e.g., by-laws), as the fear of harassment remains a barrier to women's mobility and access to public transport. Although the Constitution does not make explicit reference to the right to transport, access to safe public transport is intrinsically linked to the right to freedom and security of the person, and the right to freedom of movement (1).

Future Research

In this research, travel distances were not explored, as these are not included in the SANHTS (9). This is in line with the way data is gathered by Statistics South Africa on behalf of the National Department of Transport. In the future it may be useful to define different categories of journey length, such as journeys less than 5 km, journeys between 5 km and 10 km, and journeys of more than 10 km, or even journeys from one urban center to another or across provincial boundaries.

Future research can look at the degree to which institutional constraints relating to the health service affect mobility patterns relating to the care of the sick. For example, recent government initiatives in South Africa, such as mobile clinics, have the potential to affect

decisions relating to transport. The newly proposed National Health Initiative may have significant effects on mobility trends relating to care.

While this research builds on international findings demonstrating that males experience less harassment than females, expanding the range of focus groups to include a more equal distribution of male and female groups may provide interesting data and enable a more nuanced understanding of both female and male choices regarding transport relating to care.

As the sample focus groups in this research were all situated within the Cape Town metropolitan area, it may be useful to compare transport-user behavior in different South African urban centers over similar distances and to contrast this with transport-user behavior in rural areas.

This research has not distinguished between the terms “violence” and “harassment” although international literature suggests these may be examined as distinct areas. A finer distinction between these terms in the language of the interview questions and responses may show increased differences of perception of the likelihood of experiencing either violence or harassment by males and females.

Acknowledgments

The authors thank Heather Allen for including them in the original work conducted for the FIA Foundation that resulted in the publication, *Safe and Sound – International Research on Women’s Personal Safety on Public Transport* (26). Their invitation to work with Heather Allen sparked a wide interest in this topic in the South African context. The authors also thank all the students who have assisted this study, in particular during the data collection stage. Without the contributions of Jennifer Baufeldt, Rea Molefi, and Felix Martinez this publication would not have been possible. Last but not least, the authors acknowledge the reviewing efforts of Cheryl Wright.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: MJWAV; data collection: MJWAV; analysis and interpretation of results: MJWAV, SRP; draft manuscript preparation: MJWAV, AJG-E, SRP. All authors reviewed the results and approved the final version of the manuscript.

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- The Standing Committee on Women's Issues in Transportation (ABE70) peer-reviewed this paper (19-03432).*